ASSIGNMENTS – OOADI

LECTURE 4

1. Create an array with 100 bicycles – each with individual characteristics. Use the Random class to generate some random attribute values, see https://www.geeksforgeeks.org/generating-random-numbers-in-java/
   Add an attribute name to the bicycle class, and code a namegenerator such that you get a random and unique name for each bicycle, e.g. "myBicycle_1".
2. Place the 100 bicycles in a LinkedList, Stack and PriorityQueue and do the following
   a. Add them one-by-one to a stack. Then decrease the speed for each bicycle by 25% as you take them off the stack.
   b. Add them one-by-one to a PriorityQueue. Then increase speed for each bicycle by 25% in the same order as you inserted the bicycles.
   c. Structure the bicycles in a LinkedList, then remove all bicycles that drives less than 10 km/h.
3. Use a hashmap to organize the remaining bicycles and add a possibility to input a name from the user. Use the inputted name to search and find the hashmapped bicycle and print its info. Repeat this until the user types "exit". To get user input you can use the Scanner class, see e.g. https://www.w3schools.com/java/java_user_input.asp